WORKED SOLUTION

STEP I 1997, Question 3

Pure • Powers and Units Digits

STEP I 1997, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s1-q3

The key idea

Before anything else, write a_7 as a single power of 3: cubing a power of 3 just triples the exponent. After that, each part asks a different question about one enormous number. The last digit depends only on the exponent modulo 4, because the last digits of powers of 3 repeat with period 4. The size bound needs only a crude comparison with a power of 10. The final part rests on that size bound: a huge a_7 makes $1/a_7$ negligible.

Solution

First we find a closed form. If $a_n = 3^k$ then $a_{n+1} = (3^k)^3 = 3^{3k}$, so the exponent is multiplied by 3 at each step. Starting from $a_1 = 3^1$, induction gives

$$a_n = 3^{3^{n-1}}, \qquad \text{so} \qquad a_7 = 3^{3^6} = 3^{729}.$$

Part (i)

The units digit of a product depends only on the units digits of its factors. So we can track the last digit of 3^k by multiplying the previous last digit by 3 and keeping only the last digit:

k	1	2	3	4	5	$\cdots$
units digit of 3^k	3	9	7	1	3	$\cdots$

Once we are back to 1 at $k = 4$, the pattern $3, 9, 7, 1$ repeats with period 4. (Put another way, $3^4 = 81 \equiv 1 \pmod{10}$, so $3^{4m+r} \equiv 3^r \pmod{10}$.)

Ask yourself How do we find 729 modulo 4 without long division?

Since $3 \equiv -1 \pmod 4$, we have $3^6 \equiv (-1)^6 = 1 \pmod 4$. (Directly: $729 = 4 \times 182 + 1$.) Hence $3^{729} = 3^{4 \cdot 182 + 1}$ has the same units digit as 3^1, and

the units digit of a_7 is 3.

Part (ii)

Group the exponent in threes: $729 = 3 \times 243$, so

$$a_7 = 3^{729} = \left(3^3\right)^{243} = 27^{243} > 10^{243} \geq 10^{100},$$

using $27 > 10$ (and the fact that raising positive numbers to a positive power preserves order). Hence $a_7 \geq 10^{100}$, as required. The bound is far from tight: in fact a_7 has 348 digits.

Part (iii)

Split the fraction:

$$\frac{a_7 + 1}{2a_7} = \frac{1}{2} + \frac{1}{2a_7}.$$

Since $a_7 > 0$, the second term is positive. By part (ii) it is also tiny:

$$0 < \frac{1}{2a_7} \leq \frac{1}{2 \times 10^{100}} < 10^{-100}.$$

Therefore

$$0.5 < \frac{a_7 + 1}{2a_7} < 0.5 + 10^{-100} < 0.505.$$

Every number strictly between 0.5 and 0.505 rounds to 0.50 to two decimal places: the third decimal place is 0, well short of the 5 that would round up. So

$$\frac{a_7 + 1}{2a_7} = 0.50 \quad \text{to two decimal places.}$$

In fact the decimal expansion begins $0.5000\ldots$, with at least 100 zeros after the 5.

Checks

A computer handles 3^{729} exactly. Computing it by cubing 3 six times confirms $a_7 = 3^{729}$, a number with 348 digits whose last digit is 3, and that $27^{243} = 3^{729} > 10^{243}$. Using exact fractions confirms that $\frac{a_7+1}{2a_7}$ lies strictly between $\frac{1}{2}$ and $\frac{1}{2} + 10^{-100}$. On a smaller scale you can test the method of part (i): $a_2 = 3^3 = 27$ has exponent $3 \equiv 3 \pmod 4$, which predicts units digit 7, and it does.

Common slips

- Reading a_7 as $3^{3 \cdot 7}$ or 3^{18} rather than 3^{3^6}: each step *multiplies* the exponent by 3, not adds 3.
- Reducing the exponent modulo 10 instead of modulo 4. The units digits cycle with period 4, so it is the exponent modulo 4 that matters.
- In part (iii), writing "0.5" with no justification. "Correct to two places" means 0.50, and you need the bounds showing that the value cannot reach 0.505.

Beyond STEP

Part (i) is a first taste of modular arithmetic and Euler's theorem: $3^4 \equiv 1 \pmod{10}$ because $\varphi(10) = 4$. Repeated use of this reduction, tower by tower, is how number theorists find the last digits of huge power towers such as $3^{3^{3^3}}$. Davenport's *The Higher Arithmetic* develops congruences and the order of an element from scratch.

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